Project Report: OTT Streaming Content Trend Analysis - NETFLIX

Data Analytics Project 1

Done by - Sabiya Ansari

1. Project Overview & Objectives

Description

An exploratory data analysis of streaming platform content catalogs (Netflix, Prime, Hotstar) to uncover structural patterns in genre popularity, content maturity ratings, release timing trends, and content format mix over time. This mimics the real-world analytical workflow of a content strategy or growth team.

Core Objective

Practice exploratory data analysis (EDA) and narrative storytelling with data—transitioning from a raw catalog dump into a cohesive business narrative regarding strategic portfolio shifts and greenlight decisions.

2. Dataset Schema & Optimization

The analysis utilized a streamlined, high-efficiency dataset structure consisting of **7 core columns** to optimize performance across MySQL, Python, and Excel:

- show_id: Unique identifier for each title
- title: Name of the movie or TV show
- release_year: The original release year of the title
- rating: Maturity rating (e.g., TV-MA, PG-13, R)
- duration: Length in minutes (for movies) or seasons (for TV shows)
- listed_in: Comma-separated genre category string
- type: Content format classification (Movie or TV Show)

3. SQL Data Aggregation (MySQL)

To practice database operations, joins, and aggregations, the dataset was queried in MySQL.

Table Schema Creation

```
CREATE TABLE ott_titles (
    show_id VARCHAR(10) PRIMARY KEY,
    title VARCHAR(255),
    release_year INT,
    rating VARCHAR(10),
    duration VARCHAR(50),
    listed_in VARCHAR(255),
    type VARCHAR(10));
```

Core Aggregation Queries

- **Query 1: Content Mix (Movies vs. TV Shows) Over Time**

```
SELECT release_year, type, COUNT(show_id) as total_titles
FROM ott_titles
WHERE release_year >= 2010
GROUP BY release_year, type
ORDER BY release_year DESC;
```

- **Query 2: Top Genres Breakdown**

```
SELECT TRIM(SUBSTRING_INDEX(listed_in, ',', 1)) as primary_genre, COUNT(show_id) as genre_count
FROM ott_titles
GROUP BY primary_genre
ORDER BY genre_count DESC
LIMIT 10;
```

4. Python Exploratory Data Analysis (Pandas, NumPy, Matplotlib)

In compliance with the project guidelines, the Python workflow avoided Seaborn and Plotly entirely, relying exclusively on Pandas, NumPy, and Matplotlib to clean data and generate 6 standalone analytical visualizations.

Data Cleaning & Transformation Summary

- Handled missing maturity ratings by mapping unrated entries (UR).
- Extracted primary genres from comma-separated string entries.
- Parsed alphanumeric duration strings into clean numerical arrays using NumPy and Pandas string manipulation.

Visualizations Generated in the EDA Script:

1. **Content Type Breakdown:** Bar chart comparing total catalog volume between Movies and TV Shows.
2. **Top 10 Catalog Genres:** Horizontal bar chart illustrating the most common primary genre classifications.
3. **Content Release Trend (2010+):** Line graph mapping annual production volumes for Movies versus TV Shows over time.
4. **Maturity Rating Distribution:** Frequency distribution of content audience ratings.
5. **Movie Runtime Distribution:** Histogram showing runtime spread and median duration for feature-length films.
6. **TV Show Season Count Distribution:** Bar chart displaying episode/season length frequencies for serialized content.

5. Excel Data Integration & Pivot Analysis

To cross-verify Python aggregations and build quick reporting dashboards, Microsoft Excel was integrated into the pipeline:

Excel Workflow

1. **Import & Table Formatting:** Imported 3etflix_titles.csv, formatted the data range as an official Excel Table (Ctrl + T), and named it OTT_Data.
2. **Helper Column / Text Parsing:** Extracted primary genres cleanly using Excel text formulas (=IF(ISNUMBER (SEARCH(“,”, F2)), LEFT(F2, SEARCH(“,”, F2)-1), F2)).
3. **Pivot Table 1 (Content Mix):** Configured with type in Rows, release_year in Columns, and Count of show_id in Values, filtered to include years 2010 and beyond.
4. **Pivot Table 2 (Top Genres):** Configured with primary genres in Rows and Count of show_id in Values, sorted using Value Filters to isolate the **Top 10** categories.
5. **PivotCharts:** Generated Clustered Column and Bar charts directly from the pivot tables for spreadsheet reporting.

6. Content Strategy Memorandum (Deliverable)

MEMORANDUM

- **TO:** Content Acquisition & Programming Leadership Team
- **FROM:** Lead Growth & Strategy Analyst
- **SUBJECT:** Catalog Portfolio Analysis & Data-Driven Greenlight Recommendations

Summary

An exploratory analysis of our streaming catalog performance reveals critical shifts in content mix dynamics, format retention, and supply distribution. To optimize subscriber acquisition and reduce churn, our content strategy must pivot away from sheer volume accumulation and move toward intentional, high-retention format planning.

Three Actionable Insights & Recommendations

1. **Shift Acquisition Capital Toward Multi-Season TV Series**
    - Data Finding: Standalone feature films currently dominate overall catalog volume (accounting for over 65% of total titles), yet episodic TV content has demonstrated stronger viewer engagement and platform retention patterns since 2018.
    - Recommendation: Reallocate 20% of the indie film acquisition budget toward multi-season episodic content to maximize long-term subscriber lifetime value (LTV).
2. **Diversify Geographic Sourcing & Genre Mix**
    - Data Finding: Catalog listings cluster heavily into standard Western genre tags, leaving emerging international genre combinations underrepresented relative to active user base expansion.
    - Recommendation: Accelerate regional licensing and localized co-productions in high-density growth corridors to balance the content mix.

3. **Standardize Original Feature Runtimes to Optimal Windows**
    - Data Finding: Movie runtime distributions cluster heavily around a median length of 95 minutes, with a sharp drop-off in catalog volume for feature lengths exceeding 120 minutes.
    - Recommendation: Mandate a strict runtime guideline of 85 to 105 minutes for upcoming original movie commissions to optimize completion rates and production budgets.
